“HR Data Analysis”

HR Data Analysis

1.Project Overview: This project is designed to analyze employee attrition patterns and workforce demographics using data from 1,470 employees. The objective is to help HR teams identify key factors driving employee turnover and enable data-driven retention strategies to improve organizational stability and performance.

2. Dataset Summary

- Rows: 1470

-Columns: 41

❖ Key Features:

Attrition, Business Travel, CF_age band, CF_attrition label, Department, Education Field, emp no, Employee Number, Gender, Job Role, Marital Status, Over Time, Over18, Training Times Last Year, Age, CF_current Employee, Daily Rate, Distance From Home, Education, Employee Count, Environment Satisfaction, Hourly Rate, Job Involvement, Job Level, Job Satisfaction, Monthly Income, Monthly Rate, Num Companies Worked, Percent Salary Hike, Performance Rating, Relationship Satisfaction, Standard Hours, Stock Option Level,

Total Working Years, Work Life Balance, Years At Company, Years In Current Role, Years Since Last Promotion, Years With Curr Manager

- Missing Data: In this dataset has no any missing value.

Data Analysis using SQL + Python + Excel + Power Bi

3. Exploratory Data Analysis (EDA) using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure or Data types and .describe() for summary statistics.

	Employee Number	Training Times Last Year	-2	0	Age	CF_current Employee	Daily Rate	Distance From Home	Employee Count	Environment Satisfaction	...	Performance Rating
count	1470.000000	1470.000000	1470.0	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	...	1470.000000
mean	1024.865306	2.799320	-2.0	0.0	36.923810	0.838776	802.485714	9.192517	1.0	2.721769	...	3.153741
std	602.024335	1.289271	0.0	0.0	9.135373	0.367863	403.509100	8.106864	0.0	1.093082	...	0.360824
min	1.000000	0.000000	-2.0	0.0	18.000000	0.000000	102.000000	1.000000	1.0	1.000000	...	3.000000
25%	491.250000	2.000000	-2.0	0.0	30.000000	1.000000	465.000000	2.000000	1.0	2.000000	...	3.000000
50%	1020.500000	3.000000	-2.0	0.0	36.000000	1.000000	802.000000	7.000000	1.0	3.000000	...	3.000000
75%	1555.750000	3.000000	-2.0	0.0	43.000000	1.000000	1157.000000	14.000000	1.0	4.000000	...	3.000000
max	2068.000000	6.000000	-2.0	0.0	60.000000	1.000000	1499.000000	29.000000	1.0	4.000000	...	4.000000

Performance Rating	Relationship Satisfaction	Standard Hours	Stock Option Level	Total Working Years	Work Life Balance	Years At Company	Years In Current Role	Years Since Last Promotion	Years With Curr Manager
1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
3.153741	2.712245	80.0	0.793878	11.279592	2.761224	7.008163	4.229252	2.187755	4.123129
0.360824	1.081209	0.0	0.852077	7.780782	0.706476	6.126525	3.623137	3.222430	3.568136
3.000000	1.000000	80.0	0.000000	0.000000	1.000000	0.000000	0.000000	0.000000	0.000000
3.000000	2.000000	80.0	0.000000	6.000000	2.000000	3.000000	2.000000	0.000000	2.000000
3.000000	3.000000	80.0	1.000000	10.000000	3.000000	5.000000	3.000000	1.000000	3.000000
3.000000	4.000000	80.0	1.000000	15.000000	3.000000	9.000000	7.000000	3.000000	7.000000
4.000000	4.000000	80.0	3.000000	40.000000	4.000000	40.000000	18.000000	15.000000	17.000000

4.KPI's Report:

- **Total Employees:** The company has a total workforce of 1,470 employee
- **Attrition Count:** A total of 237 employees has left the company.
- **Average Age:** The average age of employees is 36.92 years (~37 years).
- **Active Employees:** Currently, 1,233 employees are actively working in the organization.
- **Attrition Rate:** The overall employee attrition rate stands at 16.12%.

KPI's				
Total Employee	Attrition Count	Average Age	Active Emp	Attrition Rate
1470	237	37	1233	16.12%

4.Deep Drive Analysis:

Employees Gender Wise:

Gender	Employee Count
Female	588
Male	882

- The company has **588 female** and **882 male** employees, indicating a relatively balanced workforce, though male employees represent a higher proportion of the total staff.

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❖ Total Employees in various Department Gender Wise

- The HR department has **20 female** and **43 male** employees, while the R&D department consists of **379 female** and **582 male** employees. Similarly, the Sales department includes **189 female** and **257 male** employees, indicating a higher male representation across all departments.

Department	Gender	Employee Count
HR	Female	20
	Male	43
R&D	Female	379
	Male	582
Sales	Female	189
	Male	257

❖ Total Employees in various Education Field Gender Wise:

- Most of the male emp (366 emp.) whose Education Field is life Science. And most of the Female emp (240 emp) Whose Education field Life science.

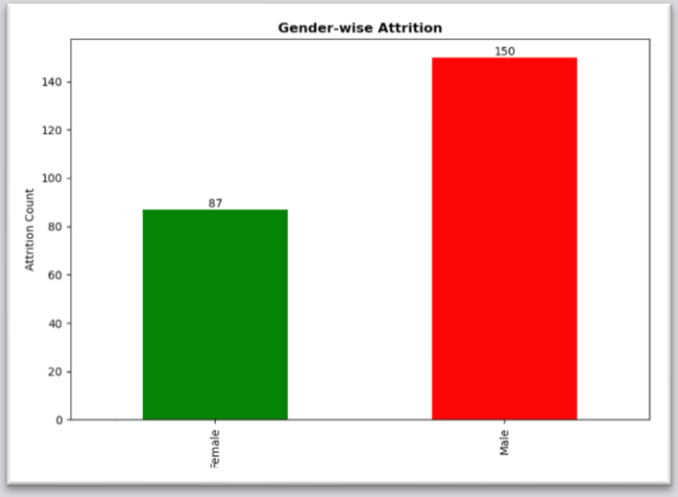
Education Field	Gender	Employee Count
Human Resources	Female	8
	Male	19
Life Sciences	Female	240
	Male	366
Marketing	Female	69
	Male	90
Medical	Female	190
	Male	274
Other	Female	29
	Male	53
Technical Degree	Female	52
	Male	80

ATTRITION

❖ Gender Wise Attrition:

Result Grid | Filter Rows:

gender	count_attrition
Female	87
Male	150



- We Observed that most of attrition by Male Emp. (150).
- Where the attrition by the male category is 150 and Female category are 87.

Attrition count by Department:

- Most the attrition in the dept of Research and Development (56.1%) And then Sales (38.8%) and HR Dept (5.1%).

Department	count_attrition	attrition_percentage
R&D	133	56.12
Sales	92	38.82
HR	12	5.06

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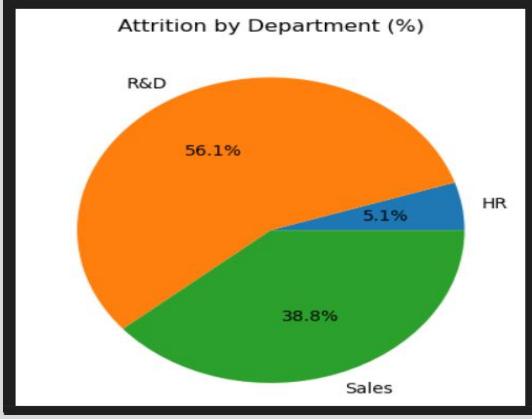
Department
HR        5.06
R&D      56.12
Sales    38.82
dtype: float64
```



❖ Attrition Count by Job Role:

- Most attrition occurs in the roles of **Laboratory Technician (62 employees)**, followed by **Sales Executive (57 employees)**, **Research Scientist (47 employees)**, and **Sales Representative (33 employees)**, indicating these positions experience the highest employee turnover.
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Job Role
Research Director              2
Manager                        5
Healthcare Representative      9
Manufacturing Director        10
Human Resources               12
Sales Representative          33
Research Scientist            47
Sales Executive               57
Laboratory Technician         62
dtype: int64
```
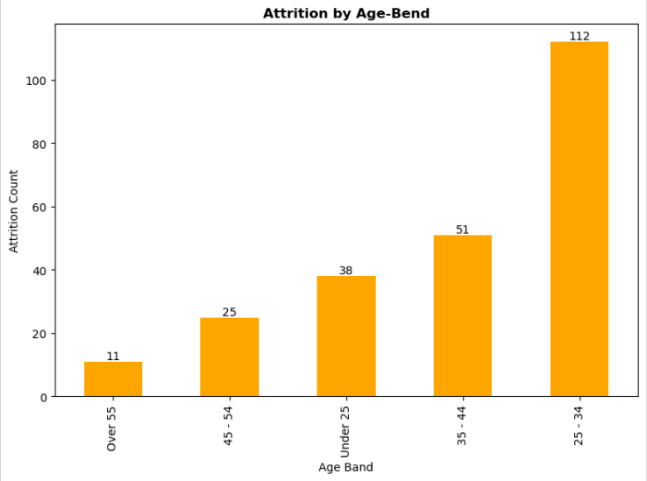

❖ Attrition By Age Band:

- The highest attrition is observed in the **25–34 age group (112 employees)**, followed by **35–44 (51 employees)** and **under 25 (38 employees)**, indicating that early to mid-career professionals are more likely to leave the organization.

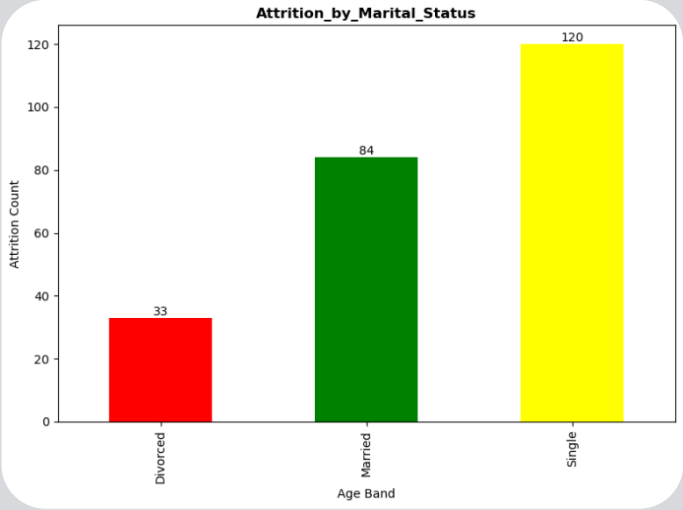


Result Grid | Filter Rows: | Export:

age_band	gender	attrition_count	attrition_pct
25 - 34	Female	43	18.14
25 - 34	Male	69	29.11
35 - 44	Female	14	5.91
35 - 44	Male	37	15.61
45 - 54	Female	9	3.80
45 - 54	Male	16	6.75
Over 55	Female	3	1.27
Over 55	Male	8	3.38
Under 25	Female	18	7.59
Under 25	Male	20	8.44

❖ Attrition By Marital Status:

Marital Status	
Divorced	33
Married	84
Single	120



- Employees with **Single marital status (120 employees)** show the highest attrition, indicating that unmarried employees are leaving the organization more frequently than others.

Job Satisfaction

❖ Average Job Satisfaction:

- The overall average job satisfaction score is **2.73 out of 4**, indicating a moderate level of employee satisfaction across the organization.

❖ Average Job Satisfaction Gender wise:

Gender	Job Satisfaction
Female	2.68
Male	2.76

- The average job satisfaction score for females is **2.68**, while for males it is **2.76**, indicating that male employees report slightly higher satisfaction levels compared to female employees.

❖ Average Job Satisfaction Department wise:

- The average job satisfaction score is **2.68** in the HR department, **2.73** in R&D, and **2.75** in Sales. This indicates that Sales employees report the highest satisfaction levels, while HR shows comparatively lower satisfaction among the three departments.

Department	Job Satisfaction
HR	2.60
R&D	2.73
Sales	2.75

❖ Average Job Satisfication by job role:

- Job satisfaction is relatively consistent across all job roles, ranging between 2.56 and 2.79, indicating no major dissatisfaction gap. However, Human Resources has the lowest satisfaction score, which may require management attention to prevent potential attrition risk.

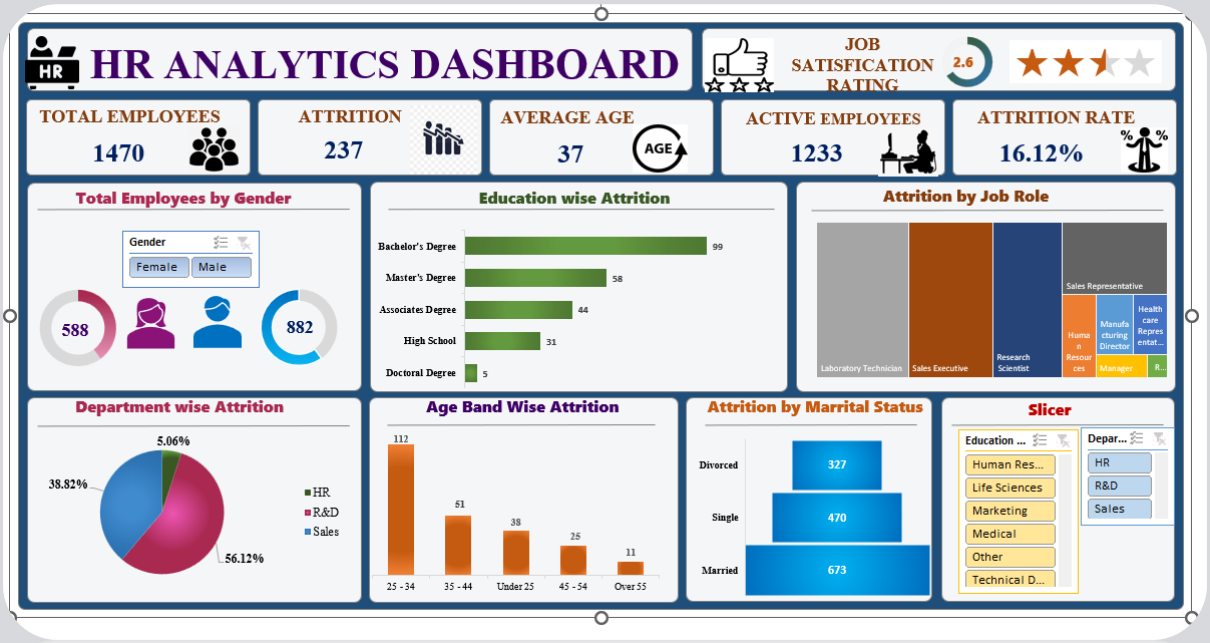
Job Role	Job Satisfaction
Healthcare Representative	2.79
Human Resources	2.56
Laboratory Technician	2.69
Manager	2.71
Manufacturing Director	2.68
Research Director	2.70
Research Scientist	2.77
Sales Executive	2.75
Sales Representative	2.73

Result Grid | Filter Rows: | Export: | Wrap

job_role	1	2	3	4	total_emp
Healthcare Representative	26	19	43	43	131
Human Resources	10	16	13	13	52
Laboratory Technician	56	48	75	80	259
Manager	21	21	27	33	102
Manufacturing Director	26	32	49	38	145
Research Director	15	16	27	22	80
Research Scientist	54	53	90	95	292
Sales Executive	69	54	91	112	326
Sales Representative	12	21	27	23	83

❖ Power Bi Dashboard:

❖ Excel Dashboard:



❖ Strategic Recommendation:

- ✓ Focus on retaining employees in high-attrition roles and those in the early stages of their careers.
- ✓ Provide clear career growth opportunities and internal promotion pathways.
- ✓ Improve employee engagement, especially for single and mid-level employees.
- ✓ Regularly track job satisfaction across departments to identify and resolve issues early.
- ✓ Use data and predictive analysis to plan workforce retention strategies more effectively.

Thank you & Happy Learning!

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